

1. $A = x^2 + 3x - 3, B = 2x^2 + x + 3, C = -3x^2 - 3x - 3$ であるとき、次の計算をせよ。

- (1) $A - C$

答.
- (2) $A - 3C$

答.
- (3) $A - B - C$

答.
- (4) $-A - B + C$

答.
- (5) $-A + 2B - C$

答.
- (6) $2A - 2B + C$

答.
- (7) $3A + B - 2C$

答.
- (8) $2A + B - C$

答.
2. 次の 1 次方程式を解け。

(1) $-8x - 1 = -9x + 4$

答.

(2) $-3x + 1 = x + 1$

答.

(3) $-5x - 5(-6x + 8) = 5$

答.

(4) $2 - 3(9x - 5) = -6x$

答.

(5) $-5x + 3 - 2(-4x + 2) = 0$

答.

(6) $6x - \frac{4}{3} = 2x + \frac{9}{4}$

答.

(7) $\frac{3}{4}x - 1 = -4x + 2$

答.

(8) $3x - \frac{5}{4} = x + \frac{5}{4}$

答.

1. $A = x^2 + 3x - 3$, $B = 2x^2 + x + 3$, $C = -3x^2 - 3x - 3$ であるとき, 次の計算をせよ。

(1) $A - C$

答. $4x^2 + 6x$

(2) $A - 3C$

答. $10x^2 + 12x + 6$

(3) $A - B - C$

答. $2x^2 + 5x - 3$

(4) $-A - B + C$

答. $-6x^2 - 7x - 3$

(5) $-A + 2B - C$

答. $6x^2 + 2x + 12$

(6) $2A - 2B + C$

答. $-5x^2 + x - 15$

(7) $3A + B - 2C$

答. $11x^2 + 16x$

(8) $2A + B - C$

答. $7x^2 + 10x$

2. 次の 1 次方程式を解け。

(1) $-8x - 1 = -9x + 4$

答. $x = 5$

(2) $-3x + 1 = x + 1$
 $-4x = 0$

答. $x = 0$

(3) $-5x - 5(-6x + 8) = 5$
 $-5x + 30x - 40 = 5$
 $25x = 45$

答. $x = \frac{9}{5}$

(4) $2 - 3(9x - 5) = -6x$
 $2 - 27x + 15 = -6x$
 $-21x = -17$

答. $x = \frac{17}{21}$

(5) $-5x + 3 - 2(-4x + 2) = 0$
 $-5x + 3 + 8x - 4 = 0$
 $3x = 1$

答. $x = \frac{1}{3}$

(6) $6x - \frac{4}{3} = 2x + \frac{9}{4}$
両辺に 12 をかけて, $72x - 16 = 24x + 27$

答. $x = \frac{43}{48}$

(7) $\frac{3}{4}x - 1 = -4x + 2$
両辺に 4 をかけて, $3x - 4 = -16x + 8$

答. $x = \frac{12}{19}$

(8) $3x - \frac{5}{4} = x + \frac{5}{4}$
両辺に 4 をかけて, $12x - 5 = 4x + 5$

答. $x = \frac{5}{4}$